

Extra Project: Keypad Interfacing

```
/*
 * File:  main.c
 * Author: SAGAR
 *
 * Created on 8 June, 2025, 8:33 AM
 */

#include <xc.h>
#include <stdint.h>
#include <stdbool.h>
#include "LCD_Header.h"
#include "keypad.h"

#define _XTAL_FREQ 20000000

// LCD Display Labels
#define LCD_POS_NUM1 0x80 // Line 1, position 0
#define LCD_POS_OP 0x87 // Line 1, position 7
#define LCD_POS_NUM2 0x8A // Line 1, position 10
#define LCD_POS_RES 0xC7 // Line 2, position 7

// Variables
char RESULT[8] = "RESULT: "; // 7 chars + null
uint16_t num1 = 0, num2 = 0, result = 0;
char operation = 0;
bool flag = false;

void main(void) {
    OPTION_REG &= ~(0x01 << 7); // Enable internal pull-up on PORTB
    TRISB = 0xF0; // RB7-RB4 inputs (keypad rows), RB3-RB0 outputs (columns)
    PORTB = 0xFF;

    char key;
    uint8_t digit;

    Lcdinit();

    // Print second line with "RESULT:" at beginning
    LcdCommand(0xC0);
    for (int i = 0; i < 7; i++) {
        LcdData(RESULT[i]);
    }
    for (int i = 7; i < 16; i++) {
        LcdData(' ');
    }
}
```

```

while (1)
{
    key = keypad_getkey();

    if (key) {
        if (key == 'C') {
            // Reset all values and clear screen
            flag = false;
            num1 = 0;
            num2 = 0;
            result = 0;
            operation = 0;

            LcdCommand(LCD_POS_NUM1);
            for (int i = 0; i < 16; i++) LcdData(' ');

            LcdCommand(0xC0);
            for (int i = 0; i < 7; i++) LcdData(RESULT[i]);
            for (int i = 7; i < 16; i++) LcdData(' ');
        }

        // If digit pressed
        else if (key >= '0' && key <= '9') {
            digit = key - '0';

            if (!flag) {
                num1 = num1 * 10 + digit;
                LcdCommand(LCD_POS_NUM1);
                LcdOutput(num1);
            } else {
                num2 = num2 * 10 + digit;
                LcdCommand(LCD_POS_NUM2);
                LcdOutput(num2);
            }
        }

        // Operator pressed
        else if (key == '+' || key == '-' || key == 'X' || key == '%') {
            operation = key;
            flag = true;

            LcdCommand(LCD_POS_OP);
            LcdData(operation);
        }

        // Equals pressed
        else if (key == '=') {
            switch (operation) {
                case '+': result = num1 + num2; break;
                case '-': result = num1 - num2; break;
                case 'X': result = num1 * num2; break;
            }
        }
    }
}

```

```
        case '%':
            if (num2 != 0)
                result = num1 / num2;
            else
                result = 0;
            break;
    }

    LcdCommand(LCD_POS_RES);
    LcdOutput(result);
}
}
}
```

// LCD.c

#include <xc.h>

#include "LCD_Header.h"

#define _XTAL_FREQ 20000000

/* LCD output function */

void LcdOutput(uint16_t i)

{

uint8_t d1, d2, d3, d4; // Creating local var to reduce memory consumption

d3 = (uint8_t) ((i % 1000) / 100); // Extract hundreds place

d2 = (uint8_t) ((i % 100) / 10); // Extract tens place

d1 = (uint8_t) (i % 10); // Extract ones place

LcdCommand(0x8A);

LcdData(0x30 + d3);

LcdData(0x30 + d2);

LcdData(0x30 + d1);

}

// Function to initialize the LCD

void Lcdinit(void) {

TRISC = 0x00; // Set PORTC as output (for control signals)

TRISD = 0x00; // Set PORTD as output (for data signals)

__delay_ms(100); // Wait for LCD to stabilize

// LCD initialization sequence as per the HD44780 LCD datasheet

LcdCommand(0x30); // Send function set command (8-bit mode)

__delay_ms(100); // Delay for command execution

LcdCommand(0x30); // Repeat function set command

__delay_ms(100);

LcdCommand(0x30); // Repeat function set command again

__delay_ms(100);

LcdCommand(0x38); // Set LCD for 8-bit mode, 2-line display, 5x8 font

__delay_ms(100);

LcdCommand(0x0C); // Turn on display, cursor off

__delay_ms(100);

LcdCommand(0x01); // Clear the display

__delay_ms(100);

}

// Function to send data (characters) to the LCD

void LcdData(uint8_t i) {

PORTC |= (0x1 << 3); // Set RS (RC3) = 1 (indicates data mode)

```

PORTD = i;          // Place data on PORTD
PORTC |= (0x1 << 0); // Set EN (RC0) = 1 (enable pulse start)
__delay_ms(100);    // Small delay for command execution
PORTC &= ~(0x1 << 0); // Set EN (RC0) = 0 (enable pulse end)
}

// Function to send commands to the LCD
void LcdCommand(uint8_t i) {
    PORTC &= ~(0x1 << 3); // Set RS (RC3) = 0 (indicates command mode)
    PORTD = i;          // Place command on PORTD
    PORTC |= (0x1 << 0); // Set EN (RC0) = 1 (enable pulse start)
    __delay_ms(100);    // Small delay for command execution
    PORTC &= ~(0x1 << 0); // Set EN (RC0) = 0 (enable pulse end)
}

```

// keypad.c

```
#include <xc.h>
#include "keypad.h"
#define _XTAL_FREQ 20000000 // Change if using different crystal

// Keymap for your 4x4 keypad
const char keypad[4][4] = {
    {'7', '8', '9', '%'},
    {'4', '5', '6', 'X'},
    {'1', '2', '3', '-'},
    {'C', '0', '=', '+'}
};

// Function to scan and return key press
char keypad_getkey(void) {
    TRISB = 0xF0; // RB7-RB4 as input, RB3-RB0 as output
    PORTB = 0x00;

    for (char row = 0; row < 4; row++) {
        PORTB = ~(1 << row) & 0x0F; // Force only lower 4 bits to be active (rows)

        for (char col = 0; col < 4; col++) {
            if (!(PORTB & (1 << (col + 4)))) { // Check if key is pressed
                __delay_ms(20); // Debounce
                while (!(PORTB & (1 << (col + 4)))); // Wait for key release
                __delay_ms(20);

                if (row == 3 && col == 0)
                    return 'C'; // ON/C as 'C'
                return keypad[row][col];
            }
        }
    }
    return 0; // No key pressed
}
```

// LCD_Header.h

```
/*  
 * File: LCD_Header.h  
 * Author: sagar  
 *  
 * Created on 8 June, 2025, 8:47 AM  
 */
```

```
#ifndef LCD_HEADER_H  
#define LCD_HEADER_H
```

```
#ifdef __cplusplus  
extern "C" {  
#endif
```

```
/* Function prototypes */  
void Lcdinit(void);  
void LcdOutput(uint16_t i);  
void LcdData(uint8_t i);  
void LcdCommand(uint8_t i);
```

```
#ifdef __cplusplus  
}  
#endif
```

```
#endif /* LCD_HEADER_H */
```

// keypad.h

/*

* File: keypad.h

* Author: sagar

*

* Created on 8 June, 2025, 9:08 AM

*/

#ifndef KEYPAD_H

#define KEYPAD_H

#ifdef __cplusplus

extern "C" {

#endif

char keypad_getkey(void);

#ifdef __cplusplus

}

#endif

#endif /* KEYPAD_H */